

Week 40 Loop color certificates

Facilitator guide keyed to the revised student packets

Mathematical purpose

A knot is a tame closed embedded loop in three-dimensional space, considered up to ambient deformation without cutting or letting strands pass through one another. A regular knot diagram records which strand goes over at each transverse crossing. A crossing in the picture is not a junction in the cord.

Arcs and colorings. An arc runs from one underpass to the next, continuing through overpasses. A crossing-free closed loop is one whole arc. Give every arc one of R, B, G. At each crossing the over-arc and the two under-arcs must be all one color or all three different. This guide counts all valid colorings, including constants. “Nontrivial” or “nonconstant” means more than one color is used.

Exact target counts. In the final trefoil drawing, all three labeled arcs meet at every crossing. Its valid assignments are the three constant triples and all six permutations of R, B, G: nine total, six nonconstant. The plain loop has one arc and hence exactly three total colorings, all constant.

Invariant and theorem. Reidemeister moves induce bijections of coloring sets, fixing constant colorings. Together with the supplied Reidemeister theorem for tame knots, the total number is invariant under knot equivalence. The printed two-crossing move demonstrates one local bijection; it does not by itself prove the global theorem. Algebra verifying all move types is on page 6.

One-way certificate. Different coloring counts prove knots inequivalent. Equal counts do not prove equivalence. The figure-eight knot and the plain loop both have only three colorings, though the former is knotted. Failing to untangle a cord is not an impossibility proof; a successful legal untangling is evidence of equivalence for that actual model.

Band mapping and gates

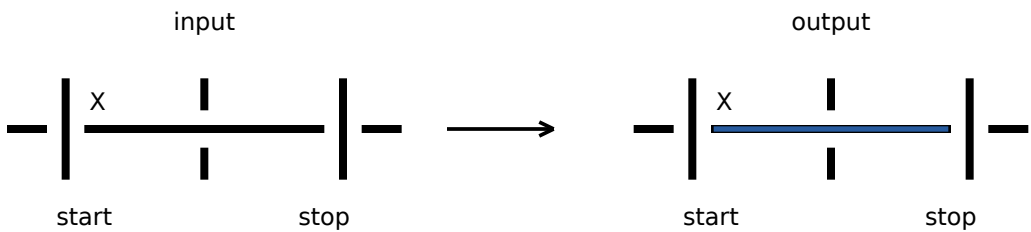
K-1: optional arc tracing, whole-arc coloring, and local missing-color cases. **Grades 2-3:** exhaust the trefoil colorings and establish unique completion in the two-crossing move. **Grades 4-5:** combine counts with the supplied invariant and explain its limit. No modular arithmetic is required of children.

Before-use gate: exact cord models, print/marker channel fit, and child handling have not been physically pretested. K-1 must distinguish over/under and trace a complete arc on the unworked target in Problem 1; if this fails, do not proceed to coloring tasks. Keep all cords on the table; children never cut or wear loops. Diagram checks do not certify the physical models.

Preparation demonstration and a flexible hour

Materials. Per table: four short, thick, flexible closed cords, about 400-600 mm each, in a tray: a plain loop, a trefoil, and two complicated unknots. Per child: final packets, tracing paper, and R/B/G markers paired with letters or distinguishable patterns. The target diagrams are 148 mm wide; do not shrink them. The final 2.4 mm coloring channel and underpass gaps still need a test with your actual markers.

Adult model preparation and required physical test. Make the two complicated unknots by visibly reversible motions from a simple closed loop, such as adding a curl and lifting a bent strand over another segment. Record and reverse your sequence. For the trefoil, form a loose overhand knot in an open cord and securely join its ends without introducing crossings, then lay it out to match the printed three-crossing model. Verify the whole closed cord, every over/under relation, and the join. A photograph or generated curve does not perform this check. Reject damaged cords, exposed wire, tight knots, electrical cords, and unsafe small closures.



Launch, 6-8 minutes. Let children handle the tray briefly. In the matching X example above and on student page 1, start just after the left underpass, follow the horizontal strand over the middle crossing, and stop just before the right underpass. Do not switch to the vertical strand. Then point to the continuous overstrand in the three isolated examples: R/R/R and R/B/G are valid; R/R/B is not. Leave the target trefoil uncolored.

Time	Suggested work
0-8 min	Handle cords, show an arc, and check each child’s transfer.
8-24 min	Problems 2-3: trefoil and plain-loop colorings.
24-36 min	Problem 4: actual attempts with the four checked cord models.
36-51 min	Local crossing/completion problems appropriate to the band.
51-60 min	Compare experiments with certificates and discuss the one-way limit.

Readiness response. Ask the child to trace one complete unworked target arc before coloring. If the child switches strands or stops at every visible crossing, return to the physical overpass example. K-1 participation in later color tasks is conditional, not automatic. All cutting and material preparation are adult-only.

K-1 answer key

Problem 1. In each isolated crossing, the continuous vertical strand is over. In the X example, the horizontal arc passes over the middle strand and under the end strands. Accept any complete labeled target arc traced gap-to-gap through its overpass. Check this transfer before proceeding.

Problem 2. Reading a record as (arc 1, arc 2, arc 3), the complete set is RRR, BBB, GGG, RBG, RGB, BRG, BGR, GRB, GBR. There are nine, not twelve; the extra recording places do not announce the answer. Exactly two equal arc colors would put “two same and one different” at every crossing and are invalid.

Problem 3. No. The plain loop has only one whole arc, so its entire length must have one color. Its three possible colorings are all red, all blue, or all green.

Problem 4. For correctly prepared models, the plain loop and two complicated unknots can become the plain shape; the trefoil cannot. Record what actually happens. A stuck cord does not prove impossibility; an unexpectedly untangled model may have been assembled incorrectly. No global proof is expected of this band.

Problem 5. In the printed 3-by-3 arrangement, the left under-color is fixed by the row (R, B, G) and the over-color by the column (R, B, G). Missing right under-colors are:

Left under-color	Over R	Over B	Over G
R	R	G	B
B	G	B	R
G	B	R	G

No question mark has two answers. If the known colors agree, the third must agree. If they differ, the third must be the remaining color.

Problem 6. Cases are numbered in reading order, left-to-right then top-to-bottom. L,P,R,Q are endpoint colors, not new names for separate arcs.

Case	(L,P,R,Q)	Works	Middle horizontal color
1	(R,B,R,B)	Yes	G
2	(R,R,B,R)	No	Second crossing forces right R, not B
3	(G,B,G,B)	Yes	R
4	(B,G,B,R)	No	P and Q disagree on one whole arc
5	(B,B,B,B)	Yes	B
6	(R,G,R,G)	Yes	B

Successful cases have P=Q and L=R. The unique middle color follows the Problem 5 table; passing under the same color twice restores the starting color.

Grades 2-3 answer key and target verification

Problem 1. Identify the unbroken vertical strand as the overstrand in each isolated example. On the target, follow a labeled arc through its overcrossing, stopping at its two underpass gaps. The colored X example teaches this representation; it is not a coloring of the target.

Problem 2. Arc-order triples: RRR, BBB, GGG, RBG, RGB, BRG, BGR, GRB, GBR. All nine satisfy all three crossings. To show completeness, pick colors for arcs 1 and 2. If they agree, arc 3 is forced to agree; if they differ, arc 3 is forced to be the remaining color. There are $3 \times 3 = 9$ choices of the first two colors, each with exactly one completion.

Crossing in final drawing	Over-arc	Two under-arcs
Upper left	3	1 and 2
Upper right	2	3 and 1
Lower center	1	2 and 3

This is the actual final crossing/arc relation, checked against the generated spatial trefoil and visible gaps. It matters more than the silhouette: reversing one crossing can change the knot. The three printed numbers label whole arcs, not regions enclosed by the curve.

Problem 3. Exactly three colorings, one constant choice each of R, B, G; none uses more than one color because the crossing-free loop is a single arc.

Problem 4. Expected for correctly assembled models: the three unknots can be legally opened into the plain shape; the trefoil cannot. Successful legal motions establish what was achieved. Failure within a short session leaves the question unresolved by experiment. A formal trefoil obstruction additionally uses the invariant, not frustration or elapsed time.

Problem 5. All nine ordered choices of (L,P) work: (R,R), (R,B), (R,G), (B,R), (B,B), (B,G), (G,R), (G,B), (G,G). Both pictures have R-end color equal to L and Q-end color equal to P. In the before picture, the middle horizontal color is L if $L=P$ and the third color otherwise. In the after picture, the whole horizontal strand is L and the bent strand is P.

Problem 6. No. For fixed $L=a$ and $P=b$, the first crossing uniquely determines the middle horizontal color c . The second crossing with the same over-color b uniquely forces the right color back to a . The bent strand is one arc so $Q=b$. The after picture is also uniquely colored by its two strands. This gives a bijection preserving all four endpoint colors, not just a comparison of how many examples happened to work.

Grades 4-5 answer key

Problem 1. Use the same over/under and whole-arc transfer check as the other bands. A child who colors only the visible segment between all projected crossings has not yet represented the coloring object correctly.

Problem 2. Nine total colorings: RRR, BBB, GGG and the six permutations RBG, RGB, BRG, BGR, GRB, GBR in labeled arc order (1,2,3). Three are constant and six nonconstant. Completeness follows from the first-two-arcs argument on page 4; every crossing involves all three arcs, so a choice valid at one is valid at the others.

Problem 3. The plain loop has exactly three colorings, since it has exactly one arc. Thus the totals differ: 9 versus 3. Do not count rotations of the paper or permute the arc labels as an equivalence relation; the worksheet uses labeled arcs and fixed named colors.

Problem 4. State observations separately from conjectures. Three correctly built unknots admit a successful deformation to the plain shape. Inability to do this for the fourth in class is only a failed attempt. The count argument in Problem 6 supplies the impossibility certificate for the verified trefoil.

Problem 5. Every ordered pair (L,P) from R/B/G works, and each has one completion in each picture. Write $L=a$, $P=b$, and identify colors with 0,1,2 modulo 3. At the first crossing the middle under-color is $c=2b-a$. At the second it becomes $2b-c=a$. The continuous bent over-arc gives $Q=b$; hence $R=a$. In the after picture those are the two whole-strand colors. The nine-case table on page 3 is the same operation without modular notation.

Problem 6. The three-crossing loop cannot become the plain loop by the permitted deformation. If it could, the supplied invariant would force equal coloring counts. But the verified target has 9 and the plain loop 3. This is a contradiction, so the deformation is impossible. The theorem is supplied; the children have checked one local move, not proved that every possible deformation decomposes into such moves.

Problem 7. Equal counts only mean this invariant has not distinguished the two loops. They may be equivalent or inequivalent. The provided figure-eight/unknot pair demonstrates the failure of the converse. Neither “three colorings” nor “no nonconstant coloring” proves that a knot is an unknot.

Optional hints

For exhaustive coloring: “Choose the colors of two labeled arcs; what is left for the third?” For the local move: “Does the bent strand ever pass under anything?” For the certificate: “If a legal deformation existed, what number would have to stay the same?” Never change an over/under crossing just to make the coloring fit.

The full invariant and before-use record

Adult proof of the local algebra

Encode colors in the field with three elements. With incoming under-color a and over-color b , the unique outgoing under-color is $a*b=2b-a$ modulo 3. This is equivalent to the printed all-same/all-different rule. The operation satisfies the three Reidemeister identities:

Type I: $a*a=a$.

Type II: $(a*b)*b=2b-(2b-a)=a$.

Type III: $(a*b)*c=a-2b+2c$, while $(a*c)*(b*c)=2(2c-b)-(2c-a)=a-2b+2c$.

These identities and the unique crossing completion give invertible local extensions with unchanged colors outside the move disk. The inverse move recovers the original coloring, so the correspondence is a bijection. Constant colorings correspond to themselves. Reidemeister's theorem then supplies the passage from local moves (with planar isotopy) to all equivalences of tame knots. Therefore total and nonconstant coloring counts are invariants. This argument does not make matching counts a complete test.

Primary sources and precise use

Mladen Bestvina, *Knots: a handout for Summer Math Program* (2007), §§1-4, especially §2 on Reidemeister moves and §3 on tricolorability. For the figure-eight counterexample, §4, Problem 11 (PDF p. 8) gives n -colorability exactly when 5 divides n , ruling out nonconstant 3-colorings and supplying a nontrivial 5-coloring obstruction to the unknot. [Handout PDF](#).

Louis H. Kauffman, *Knots*, "The Three Colored Trefoil" and "The Quandle and the Determinant of a Knot," gives the coloring obstruction and the operation $2b-a$. The target counting and endpoint-case keys here are independently enumerated from the final diagram. [Author's exposition](#).

The printed trefoil is an original rendering of $(\sin t+2 \sin 2t, \cos t-2 \cos 2t, -\sin 3t)$, with depth deciding over/under. Its three true crossings have widely separated positions and underpasses delimit three arcs. Numerical and visual diagram checks establish the intended mathematical drawing; they do not construct or certify the classroom cord.

Record the required physical checks

Date and adult: _____

All four securely closed models checked, including crossing order: _____

Both complicated unknots successfully undone by recorded legal moves: _____

100% print, marker channel, tracing paper, and cord handling tested: _____

K-1 independent arc-tracing transfer observed: _____

Status: exact-material checks above remain unperformed here. This guide and the revised packets are unpiloted. Keep loops on the table, allow no child cutting, and make no claim that a rendered diagram certifies a successful physical model.

Key alignment: final F40-K-v2 has 5 pages and Problems 1-6; F40-23-v2 has 4 pages and Problems 1-6; F40-45-v2 has 4 pages and Problems 1-7. Prepared October 3, 2026.